CIS 4374 Project Management Smart Parking Platform Homework 5 team resources, a cost plan with monthly cash flow, and a RACI matrix.

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1. Research on Existing Parking Applications

ParkMobile allows drivers to pay for parking and make reservations. It also offers management tools for parking providers. A strength is its support for both drivers and operators. However, its website does not specify a consistent live availability update across facilities.
SpotHero helps drivers compare parking locations and prices, reserve parking, and access a parking pass. Its straightforward booking process is a strength. One limitation is that modification and cancellation options depend on the reservation terms.

ParkWhiz allows drivers to compare prices, reserve parking, and prepay before arrival. Its mobile parking pass simplifies entry. However, the reviewed page does not explain how frequently occupancy information updates or how delayed information is displayed.

Our proposed difference: The Smart Parking Platform will emphasize clearly labeled availability updates, reservation management, and operator reporting. Drivers will see when availability was last updated and receive a warning when information is outdated. These are proposed priorities, rather than claims that competitors lack these features.

2. Vision and Scope

Project Background and Company Intro

For this fictional project, we assume a group of parking garage operators approached WE ARE $oftware ¢orp to plan a system that improves parking search and reservation management. WE ARE $oftware ¢orp is a fictional software development studio with unlimited theoretical resources and budget. The project must still follow the semester timeline.

Business Need

Drivers can spend unnecessary time searching for parking, causing frustration and congestion. Parking operators also need tools to monitor occupancy and manage available spaces.

The Smart Parking Platform will help drivers locate, reserve, and pay for parking through web and mobile applications. It will provide directions, reservation notifications, and receipts. Operators will use a dashboard to manage availability, prices, and occupancy reports.

Initial Scope

The proposed system includes:

- User accounts and vehicle information
- Parking search, interactive maps, and current availability
- Reservations, digital payments, history, and receipts

- Notifications and navigation assistance
- Operator tools for availability, pricing, and reporting

The initial scope excludes garage construction, sensor manufacturing, physical maintenance, and parking enforcement.

3. Initial SRS and Use Cases

Purpose and Users

This draft identifies the system's initial requirements. Drivers will use the web and mobile applications, while parking operators will use a management dashboard. Requirements will be expanded as stakeholder needs to become clearer.

Functional Requirements

The system shall allow drivers to manage accounts, find parking, make reservations, pay, and review their parking activity. It shall also allow authorized operators to manage availability and prices and view occupancy reports.

Proposed Nonfunctional Requirements

These initial targets will be reviewed later:

- Parking search results should be loaded within two seconds under an agreed test workload.
- Availability of readings older than 60 seconds should be marked as outdated.
- Users should only access records and functions permitted for their role.

Initial Use Cases

- **Register an Account**
  A new driver enters account information. The system validates the details and creates an account after verification.

- **Log In**
  A registered user submits valid credentials. The system opens the appropriate driver or operator interface.

- **Manage Profile and Vehicles**
  A logged-in driver updates contact details or vehicle information. The system validates and saves the changes.

- **Search for Parking**
  A driver enters a destination and parking period. The system displays matching garages.

- **View Parking on a Map**
  A driver opens the map and selects a garage marker. The system displays that garage's information.

- **View Current Availability**
  A driver selects a garage. The system displays reported availability and its latest update time.

- **Reserve Parking**
  A logged-in driver selects a garage, time, and vehicle. The system checks inventory and confirms the reservation after successful payment.

- **Modify a Reservation**
  A driver requests changes to an existing booking. The system checks availability and policy restrictions before saving approved changes.

- **Cancel a Reservation**
  A driver cancels an eligible booking. The system releases the reserved capacity and records any applicable refund.

- **Pay for Parking**
  A driver submits payment during checkout. The system records the payment provider's approval or failure of results.

- **View History and Receipts**
  A logged-in driver selects a past reservation. The system displays its details and provides a receipt for completed payments.

- **Receive Notifications**
  A reservation event occurs. The system sends a confirmation, reminder, or update through the driver's approved notification channel.

- **Navigate to a Garage**
  A driver selects Directions. The system opens a mapping service with the garage entrance as the destination.

- **Manage Availability and Prices**
  An authorized operator updates garage availability or rates. The system checks for conflicts and saves valid changes.

- **View Occupancy Reports**
  An authorized operator selects a garage and date range. The system displays occupancy information and identifies missing data.

4. Work Breakdown Structure

The Work Breakdown Structure divides the Smart Parking Platform into three levels: the overall platform, its major components, and the features or deliverables within each component.

1.0 Smart Parking Platform

1.1 Project Planning

1.1.1 Vision and Scope document
1.1.2 Software Requirements Specification
1.1.3 Project schedule and stakeholder updates

1.2 System Design

1.2.1 Web and mobile interface designs
1.2.2 Database and API design
1.2.3 Payment, mapping, and garage integration designs

1.3 User Accounts

1.3.1 Registration, login, and password recovery
1.3.2 Driver profiles and vehicle records
1.3.3 Operator accounts and access permissions

1.4 Parking Search and Garage Management

1.4.1 Parking search and interactive map
1.4.2 Garage details, availability, and navigation
1.4.3 Operator controls for garage information, availability, and pricing

1.5 Reservations and Payments

1.5.1 Reservation creation, modification, and cancellation
1.5.2 Payment processing and refunds
1.5.3 Reservation history and receipts

1.6 Notifications and Reporting

1.6.1 Reservation confirmations and reminders
1.6.2 Occupancy reports and graphs
1.6.3 Occupancy report exports

1.7 Testing and Release

1.7.1 Functional and integration test results
1.7.2 Security, usability, and user acceptance test results

1.7.3 Release package and user instructions

5. Draft Timeline and Gantt Chant

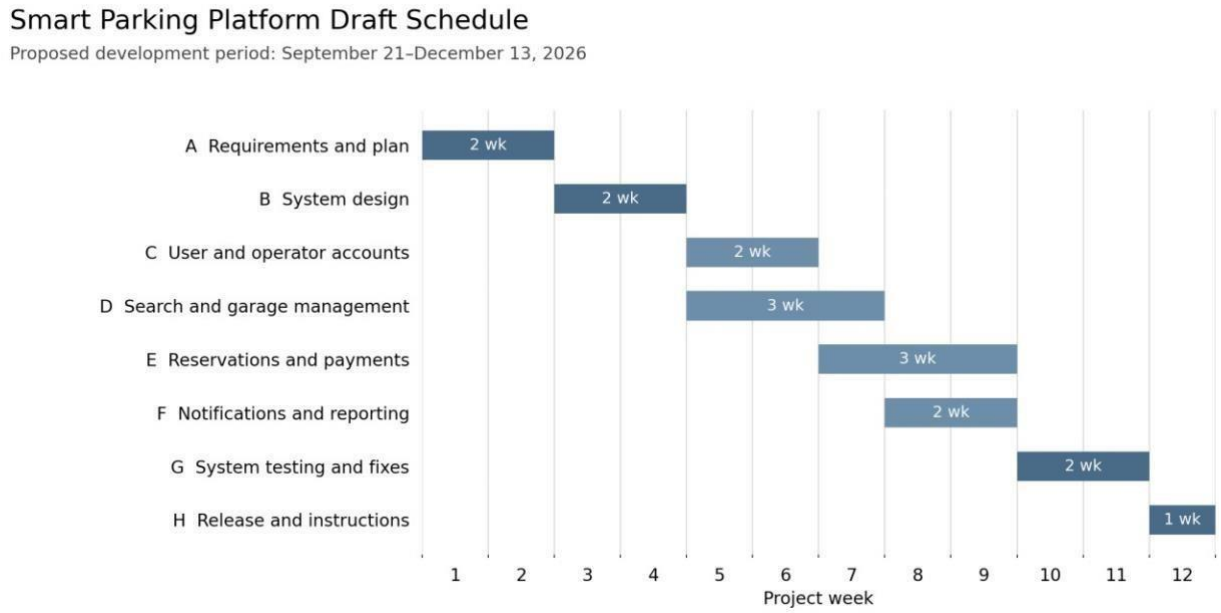
This draft assumes a 12 week development period from September 21 through December 13, 2026. The dates and durations are preliminary estimates for the fictional development project.
They are not the course submission schedule and will be revised as requirements become clearer.

Separate development teams are assumed to work on some components at the same time. Each development activity includes component testing. Project coordination, document revisions, and stakeholder updates continue throughout the schedule.

ID	Activity	WBS	Weeks long	Planned weeks	Depends on
A	Requirements and initial plan	1.1	2	1-2	None
B	Interfaces, database, and integration design	1.2	2	3-4	A
C	User and operator accounts	1.3	2	5-6	B
D	Parking search and garage managements	1.4	3	5-7	B
E	Reservation and payments	1.5	3	7-9	C

F	Notifications and reporting	1.6	2	8-9	C, D
G	System testing and defect resolution	1.7.1-1.7.2	2	10-11	E, F
H	Release and user interactions	1.7.3	1	12	G

Gantt Chart



Major Milestones

- End of Week 2: Initial requirements and project plan ready for review.
- End of Week 4: System design complete.
- End of Week 9: Planned features ready for full system testing.
- End of Week 11: Testing complete and release-blocking defects resolved.
- End of Week 12: Release package ready for handoff.

The schedule is a starting estimate. Delays in garage data access or third-party integrations may require changes to the planned dates.

6. Product Backlog and Spring 1 Planning

Product Backlog

I created a backlog in Jira to organize the work needed for the Smart Parking Platform. The backlog contains 45 tasks divided into four categories:

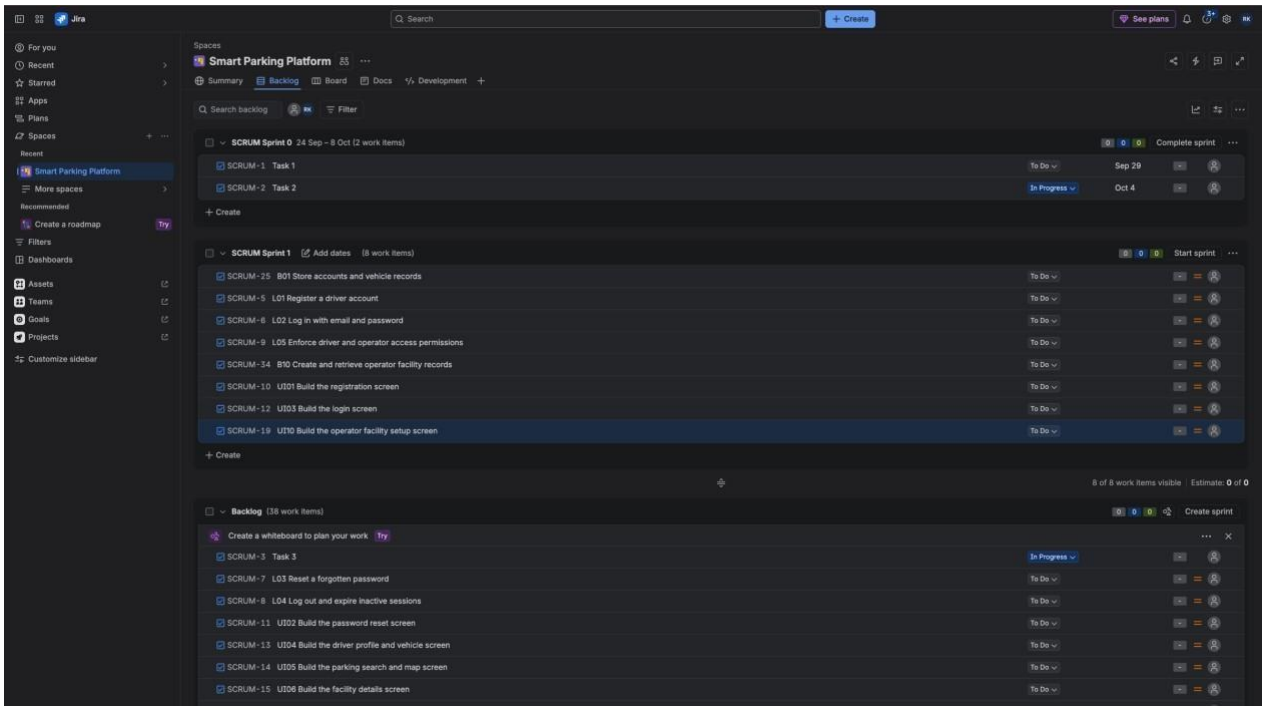
Category	Number of Tasks
Login	5
User interface for drivers and operators	15
Backend processes for drivers and operators	15
Reporting for drivers and operators	10
Total	**45**

Each task includes a description and completion criteria. The sample tasks provided by Jira are not included in these totals.

The goal of Sprint 1 is to allow drivers to register and log in, and allow authorized operators to add a parking facility. I selected the following eight tasks because they provide the account access and facility setup needed for later parking features.

Task Code	Planned Task
B01	Store accounts and vehicle records
L01	Register a driver account
L02	Log in with email and password
L05	Enforce driver and operator access permissions
B10	Create and retrieve operator facility records
UI01	Build the registration screen
UI03	Build the login screen
UI10	Build the operator facility setup screen

The remaining 37 parking tasks stay in the backlog for future sprints. All eight Sprint 1 tasks are marked To Do because this homework focuses on planning. Development has not started.



7. Project Risks

The following risks could affect the planned development of the Smart Parking Platform. These are possible future problems, not incidents that have already occurred.

Technical Risks

- **T1: Outdated parking availability.** Delayed garage updates could cause drivers to reserve parking based on inaccurate availability.
- **T2: Payment service failure.** A payment provider outage could prevent drivers from completing reservations or receiving refunds.
- **T3: Unauthorized access.** Incorrect access permissions could expose driver records or allow an operator to change another garage's information.
- **T4: Duplicate reservations.** Simultaneous booking requests could reserve more spaces than a garage has available.

Schedule Risks

- **S1: Late requirements changes.** New feature requests could require additional design and development work.
- **S2: Delayed garage data access.** Operators might not provide sample data or integration details when needed.

- **S3: Underestimated development time.** Reservations, payments, or other features could take longer to build than planned.
- **S4: Major defects found late.** Problems discovered during final testing could delay the planned release.

Financial Risks

- **F1: Higher hosting costs.** Increased usage or inefficient processing could raise cloud service costs.
- **F2: Higher service fees.** Mapping, payment, or notification providers could increase their charges.
- **F3: Additional rework costs.** Unclear requirements or design mistakes could increase development expenses.
- **F4: Payment error losses.** Duplicate charges or incorrect refunds could create financial losses and additional support work.

People Risks

- **P1: Team member unavailability.** An absence or departure could interrupt work on an important feature.
- **P2: Limited experience.** Team members might need additional training to implement payment processing or security controls.
- **P3: Poor coordination.** Teams could misunderstand responsibilities or build components that do not work together.
- **P4: Delayed stakeholder feedback.** Drivers or parking operators might not review designs promptly, allowing misunderstandings to continue.

Risk Register

This initial register tracks ten risks from Section 7. Probability and impact ratings are planning estimates and will be reviewed as the project develops.

Probability describes how likely a risk is to occur: Low, Medium, or High. Impact describes the expected effect: Low means minor rework, Medium means noticeable disruption, and High means a major effect on the schedule, essential features, finances, or user data.

ID	Description	Probability	Impact	Owner	Response Strategy	Status/ Notes

T1	Delayed garage updates cause inaccurate availability.	Medium	High	Backend Lead	Mitigate: Display the last update time, flag readings older than 60 seconds, and recheck capacity before confirming a reservation. Pause booking where reliable capacity cannot be confirmed.	Open. Validate using delayed and missing data.
T2	Payment service failure interrupts checkout or refunds.	Medium	High	Payment integration lead	Mitigate: Test provider failures, track transaction status, and prevent retries from creating duplicate charges. Keep affected bookings pending until payment status is confirmed.	Open. Test before reservation features are accepted.
T3	Incorrect permissions expose records or operator controls.	Low	High	Security lead	Mitigate: Check roles and record ownership on every protected request. Review permissions and test attempts to access another user's records.	Open. Include in account and operator testing.

T4	Concurrent requests create duplicate reservations.	Medium	High	Backend lead	Mitigate: Make the capacity check and booking update one controlled database operation. Test competing requests for the last available space.	Open. Review during reservation development.
S1	Requirements change add work after planning.	Medium	Medium	Project manager	Mitigate: Record change requests, assess their effect, and obtain approval before changing the scope. Place lower priority additions in a later release.	Open. Review during weekly planning.
S2	Garage data or integration details arrive late.	Medium	High	Project manager	Mitigate: Request information early and agree on delivery dates. Use clearly labeled sample data for development while access is pending.	Open. Check before integration work begins.

S4	Major defects discovered during final testing delay release.	Medium	High	QA lead	Mitigate: Test each component during development and test integrations early. Delay release if defects affecting payments, security, or reservations remain unresolved.	Open. Review test results each week.
F1	Hosting usage exceeds the operating cost estimate.	Medium	Medium	Infrastructure lead	Mitigate: Monitor usage, set spending alerts, and remove unused resources. Investigate unexpected increases before expanding capacity.	Open. Review costs monthly.
P1	A key team member becomes unavailable.	Medium	Medium	Project manager	Mitigate: Document work, share knowledge, and identify a backup for important responsibilities. Reassign tasks when needed.	Open. Review workload and coverage weekly.

P3	Poor coordination causes incompatible components or missed tasks.	Medium	High	Development lead	Mitigate: Agree on interface requirements, assign clear task owners, and hold regular coordination	Open. Review dependencies during team meetings.
					meetings. Record decisions in shared notes.	

Communication Plan

This communication plan describes the proposed meeting schedule and reporting methods for the fictional development project. It covers the teams assumed in the project schedule. These meetings are planned activities, not a record of meetings already held.

Meetings and Reporting

Team or Audience	Meeting Frequency	Purpose	Reporting Method	Responsible Role
Project management and team leads	Weekly, 30 minutes	Review milestones, risks, dependencies, and requested changes.	Updated schedule, risk register, and meeting notes shared through Microsoft Teams.	Project manager
User interface team	Each workday, 15 minutes	Review driver and operator screens, current tasks, and blockers.	Jira task updates and design notes in Teams.	UI lead

Backend and integration team	Each workday, 15 minutes	Coordinate accounts, garage data, reservations, and payment integrations.	Jira task updates and technical decisions recorded in shared notes.	Backend lead
Quality assurance and security team	Twice weekly, 30 minutes	Review test results, permission checks, defects, and fixes needing retesting.	Jira defect reports and a shared test summary.	QA lead
All development teams	At the start and end of each two-week sprint	Select sprint work, review results, and discuss improvements.	Sprint backlog, review notes, and demonstrations of completed work when available.	Project manager
Sponsor and parking operator representatives	Weekly progress update, with a review meeting every two weeks	Explain progress, upcoming work, risks, and decisions requiring stakeholder input.	Brief written summary, a 1–2-minute update video, and review notes.	Brief written summary, a 1–2-minute update video, and review notes.

Communication Rules

Jira will track tasks, owners, blockers, and defects. GitHub will store versioned project documents and approved project files. Microsoft Teams will support discussions and meetings. Important decisions made during discussions will be recorded in shared notes or the relevant Jira task.

The weekly stakeholder update will explain what was completed, what is planned next, and which risks or decisions need attention. Technical teams will receive detailed task and defect information, while stakeholders will receive a short explanation of progress and project effects.

Problems affecting payments, user data, or the release date will be reported to the project manager as soon as they are identified. The responsible lead will record the problem, its expected effect, and the proposed next action. The project manager will notify affected stakeholders and track the response.

8. Roles and Resources

The Smart Parking Platform requires people to manage the project, design the screens, develop features, and test the system. The following staffing plan uses fictional team members and estimated skill levels.

The plan follows the proposed development period of September 21 through December 13, 2026. A full-time assignment equals 480 hours over 12 weeks, based on 40 hours per week. Allocation percentages show each person's average commitment across the project. Their weekly workload will vary by project phase.

Fictional Team Member	**Role**	**Assumed Skill Level**	**Main Responsibilities**	**Average Allocation**	**Estimated Hours**
Maya Patel	Project Manager	Experienced	Manage the schedule, risks, budget, and stakeholder updates.	25%	120
Jordan Lee	UI Developer	Intermediate	Build driver screens and the operator dashboard.	50%	240
Alex Morgan	Backend developer	Experienced	Develop accounts, reservations, payments, and garage data integration.	75%	360
Sofia Nguyen	UX designer	Intermediate	Design screen layouts and review how users complete parking tasks.	25%	120

Chris Davis	QA tester	Intermediate	Test features, record defects, and verify fixes.	37.5%	180
Taylor Brooks	Infrastructure and security engineer	Experienced	Configure hosting, deployment, monitoring, and security controls.	25%	120
Sam Rivera	Product owner	Experienced in parking operations	Clarify business requirements, prioritize features, and approve completed deliverables.	12.5%	60
Total				250%	1,200

The total allocation equals 2.5 full-time positions spread across seven people.

For this small fictional team, some people cover multiple responsibilities from the earlier risk and communication plans. Alex serves as the backend, payment integration, and development lead. Taylor covers infrastructure and security. Jordan serves as the UI lead, and Chris serves as the QA lead.

9. Resource and Cost Plan

Estimation Method

This plan uses a bottom-up estimate. Estimated hours for each role are multiplied by an assumed hourly rate. Hosting, tools, and external service allowances are then added, followed by a contingency reserve.

All amounts are in U.S. dollars. Rates and service costs are fictional planning assumptions, not vendor quotes or actual spending. Labor rates include assumed employment overhead, so no separate overhead charge is added.

Although WE ARE $oftware ¢orp. has unlimited theoretical funding, this budget provides a baseline for reviewing costs and avoiding unnecessary spending.

Labor Costs

Role	Estimated Hours	Assumed Hourly Rate	Estimated Cost
Project Manager	120	$60	$7,200
UI Developer	240	$50	$12,000
Backend Developer	360	$60	$21,600
UX designer	120	$45	$5,400
QA tester	180	$40	$7,200
Infrastructure and security engineer	120	$60	$7,200
Product Owner	60	$45	$2,700
Total	**1,200**		**$63,300**

For example, the backend development estimate is 360 hours × $60 per hour = $21,600.

Other Costs and Total Budget

Item	Basis of Estimate	Estimated Cost
Cloud hosting and storage	Allowance for development, testing, backups, and monitoring during the project.	$1,200
Software tools and licenses	Allowance for team collaboration, design, and testing tools.	$600
External services	Allowance for mapping, notifications, and limited integration testing.	$400
Test equipment	Allowance for access to additional devices for compatibility testing.	$300

Other Costs Subtotal **$2,500**

Labor	**From Section 11.2.**	$63,300
Base Estimate	$63,300 + $2,500.	**$65,800**
Contingency reserve	15% × $65,800.	$9,870
Total Proposed Budget	$65,800 + $9,870.	**$75,670**

The contingency reserve covers uncertainty such as integration problems, additional testing, or approved rework. It does not automatically fund new features.

Existing development computers are assumed to be available. Garage construction, sensor manufacturing, and physical maintenance remain outside the project scope. Operating costs after December 13, including production payment processing and ongoing support, require a separate budget.

Monthly Budgeted Cash Flow

The table shows planned funding by calendar month. September and December cover only the portions included in the project schedule. Spending is higher in October and November because most design, development, and testing work occurs during those months.

Cost Category	**Sept. 21–30**	**October**	**November**	**Dec. 1–13**	**Total**
Labor	$6,360	$23,020	$25,760	$8,160	$63,300
Hosting and storage	$150	$400	$450	$200	$1,200
Tools and licenses	$200	$150	$150	$100	$600
External services	$0	$100	$200	$100	$400
Test equipment	$100	$50	$50	$100	$300
Total Funding Allowance	**$7,831.50**	**$27,278**	**$30,601.50**	**$9,959**	**$75,670**

The contingency amounts represent funds available if needed. They are not assumed expenses. This table is a proposed budget, not a record of actual spending.

Cost Monitoring

The project manager will compare recorded labor hours and service expenses with the budget each week. Team members will update their work records, and the infrastructure engineer will review hosting usage and billing alerts.

Potential hidden costs include additional storage, data transfers, service request charges, extra software licenses, and time spent fixing defects. The team will monitor these costs and update its forecast when assumptions change.

The project manager will document requests to use contingency funds. Changes that increase the total budget or project scope will require sponsor approval.

RACI Matrix

The RACI matrix assigns responsibility for major project activities.

- **R — Responsible:** Performs the work.
- **A — Accountable:** Approves the result and is ultimately answerable.
- **C — Consulted:** Provides advice or feedback.
- **I — Informed:** Receives progress updates.
- **A/R:** Performs the work and is also accountable for it.

Each activity has exactly one accountable role.

Role abbreviations: PM = project manager, UI = UI developer, BE = backend developer, UX = UX designer, QA = QA tester, IS = infrastructure and security engineer, and PO = product owner.

Activity	PM	UI	BE	UX	QA	IS	PO
Define requirements and scope	R	C	C	C	C	C	A
Prioritize the product backlog	R	C	C	C	C	C	A
Maintain the schedule, budget, and risk register	A/R	C	C	C	C	C	C
Design driver and	I	C	C	A/R	C	I	C

operator screens							
Build the user interface	I	A/R	C	C	C	C	I
Build accounts, reservations, and payment integrations	I	C	A/R	I	C	C	C
Configure hosting and security controls	I	C	R	I	C	A/R	I
Perform system testing and verify fixes	I	R	R	C	A/R	R	C
Review and accept business functionality	C	C	C	C	R	C	A
Deploy the approved release	C	R	R	I	C	A/R	I
Prepare stakeholder updates	A/R	C	C	C	C	C	I

These assignments apply to the proposed development team. The project manager will review them when responsibilities or staffing change.